

Advanced Web Application Security Labs

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Goup: TEAM04

Sep 25, 2026

ASSIGNMENT 1 IPV4 ADDRESSING FOUNDATIONS

Part A - Core Concepts

- 1 Define an IPv4 address and explain why the address must be unique on the same network?.**

An IPv4 address is a 32 bit logical address written as four decimal octets separated by dots, e.g 192.168.1.10 its used to identify a device on a network and the reason why address must be unique on the same network because if two devices use the same IP address, packets will be delivered to the wrong device causing an IP conflict.

- 2 State one difference between flat and hierarchical addressing. Give one example of each type?.**

Flat addressing uses one part where every address is treated equally with no structure, While hierarchical addressing is divided into network and host parts to allow routing by Network.

Example of flat addressing: MAC address - 00-1A-2B-3C-4D-5E

Example of hierarchical addressing: IPv4 address - 192.168.10.10 where 192.168.10

- 3 Compare static and dynamic IP addressing. Name the protocol normally used to assign an address dynamically?**

Static addressing is manually configured and never changes. Dynamic addressing is automatically assigned by a server and may change. Protocol used: (DHCP) Dynamic Host Configuration Protocol.

- 4 State one difference between classful and classless addressing. What does CIDR stand for?**

Classful addressing uses fixed classes A, B, C with default masks determined by the first octet, while classless addressing allows variable length subnet masks. CIDR stands for Classless Inter Domain Routing.

- 5 List the four IPv4 configuration items presented in the lesson for normal name based Internet access.?**

IP address

Subnet Mask

Default Gateway

DNS address

Part B Classful Address Identification

For each address, identify the class, write the network and host pattern, and give the default subnet mask._

IPv4 address	Class	Network and host pattern	Default subnet mask
23.14.6.9	Class A	N.H.H.H	255.0.0.0
145.80.12.200	Class B	N.N.H.H	255.255.0.0
198.51.100.27	Class C	N.N.N.H	255.255.255.0
9.200.10.1	Class A	N.H.H.H	255.0.0.0
188.44.9.72	Class B	N.N.H.H	255.255.0.0

Part C Network and Broadcast Analysis

Use the default classful mask for each address.

1. 34.72.6.19

Class A, Mask 255.0.0.0

Network address: 34.0.0.0

Broadcast address: 34.255.255.255

First usable host: 34.0.0.1

Last usable host: 34.255.255.254

Maximum usable hosts: 16,777,216

2. 150.20.7.200

Class B, Mask 255.255.0.0

Network address: 150.20.0.0

Broadcast address: 150.20.255.255

First usable host: 150.20.0.1

Last usable host: 150.20.255.254

Maximum usable hosts: 65,534

3. 203.0.113.44

Class C, Mask 255.255.255.0

Network address: 203.0.113.0

Broadcast address: 203.0.113.255

First usable host: 203.0.113.1

Last usable host: 203.0.113.254

Maximum usable hosts: 254

Part D Applied Troubleshooting

Scenario Settings:

IPv4 address: 201.110.213.28

Subnet mask: 255.255.255.0

Default gateway: 201.110.214.1

DNS server: 8.8.8.8

1. Give the network address and broadcast address for the workstation.?

Network address: 201.110.213.0

Broadcast address: 201.110.213.255

Working: Class C address, mask 255.255.255.0, host bits = 0 for network, 255 for broadcast.

2. Is the configured default gateway valid for this workstation? Explain your answer.?

No, the gateway is not valid. The workstation is on network 201.110.213.0/24 but the gateway 201.110.214.1 is on network 201.110.214.0/24. A default gateway must be in the same network as the workstation to be reachable.

3. Give one valid default gateway address. Explain why the network and broadcast addresses cannot be used as the gateway.?

Valid gateway: 201.110.213.1 (Any address from 201.110.213.1 to 201.110.213.254 excluding 201.110.213.28, e.g. 201.110.213.254 is acceptable).

Reason they cannot be used: Network and broadcast addresses cannot be assigned to devices. Network address identifies the network itself, broadcast address is used to send to all hosts on the network.

4. With the incorrect gateway, what communication may still work and what communication is likely to fail?

Still work: Local communication within the same network 201.110.213.0/24, as it uses ARP directly and does not need a gateway.

Likely to fail: Communication to other networks and Internet resources (e.g. outside 201.110.213.0/24 or to 8.8.8.8) because packets cannot be routed without a valid gateway.

5. A lab contains 25 student computers. Recommend static or dynamic addressing for the computers and name the service that would provide the addresses.

Recommendation: Dynamic addressing. For 25 computers it is more efficient, avoids conflicts and reduces administration.

Service: DHCP - Dynamic Host Configuration Protocol provided by a DHCP server or DHCP-enabled router.